TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	80.4 ± 1.85
Correlation Stability	73.9 ± 3.11
Distribution Consistency	90.0 ± 1.00
Entropy Preservation	94.0 ± 0.77
Mean Z Score Drift	0.1 ± 0.04
Categorical Fidelity	93.3 ± 1.62
Privacy Risk	52.5 ± 1.86
Data Density	100.0 ± 0.01
Coverage Score	17.8 ± 2.56
Feature Accuracy	90.3 ± 3.98
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	claude-sonnet-5-5
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	46.989	+2.243	5.0%	OK
TB	3.299	2.965	-0.334	10.1%	Warn
DB	1.486	1.379	-0.107	7.2%	OK
Alkphos	290.576	279.479	-11.098	3.8%	OK
Sgpt	80.714	76.253	-4.460	5.5%	OK
Sgot	109.911	108.281	-1.630	1.5%	OK
TP	6.483	6.714	+0.231	3.6%	OK
ALB	3.142	3.333	+0.191	6.1%	OK
A/G Ratio	0.947	1.000	+0.053	5.6%	OK
Selector	1.286	1.388	+0.102	7.9%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	80.4 ± 1.85
Correlation Stability (*2)	73.9 ± 3.11
Distribution Consistency (*3)	90.0 ± 1.00
Entropy Preservation (*4)	94.0 ± 0.77
Mean Z Score Drift (*5)	0.1 ± 0.04
Categorical Fidelity (*6)	93.3 ± 1.62
Privacy Risk (*7)	52.5 ± 1.86
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	17.8 ± 2.56

Feature Accuracy (*10)	90.3 ± 3.98
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6675213675213676	0.6863247863247863	-0.020512820512820516	
F1	0.8279091624529537	0.7284000793417571	0.7606157929152796	-0.06729336953767409	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.6957264957264957	0.6230769230769231	0.06410256410256411	*
F1	0.5587432119380917	0.7850379648215341	0.6586393762135376	0.099896164275446	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	0.6564102564102565	0.6931623931623931	0.00854700854700855	
F1	0.7953194230713111	0.736023882973625	0.7788945543574618	-0.016424868713849274	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6769230769230770	0.6717948717948719	-0.014529914529914523	
F1	0.7971299319768493	0.746323316392856	0.7642063001448259	-0.03292363183202345	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6700854700854701	0.6982905982905983	-0.00427350427350427	
F1	0.7997903571549948	0.7480714865353033	0.7905734519355601	-0.009216905219434657	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6726495726495727	0.6991452991452991	-0.006837606837606825	
F1	0.8128002697439015	0.7537455743091006	0.7857441065772546	-0.02705616316664682	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7119658119658119	0.6726495726495727	0.7102564102564102	-0.00170940170940169 23	
F1	0.8136868373366214	0.7483215317053137	0.8059099217752639	-0.00777691556135747 8	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6666666666666667	0.6914529914529914	0.01709401709401709	
F1	0.7821343746683586	0.7473191093339173	0.7829347452493135	0.00080037058095486 78	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	0.6829059829059829	0.6880341880341888	0.00341880341880341 8	
F1	0.7902503364007533	0.7570162128407147	0.7843445180668617	-0.005905818333891555	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6555555555555557	0.6461538461538461	0.6700854700854701	0.014529914529914534	
F1	0.7596963808618677	0.7210517865927571	0.766407226038741	0.006710845176873359	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
2	46	Male	9.4	5.2	268	21	63	6.4	2.8	0.8
3	36	Male	0.8	0.2	182	31	34	6.4	3.8	1.4
4	60	Male	6.8	3.2	308	404	794	6.8	3.0	0.7
5	37	Male	0.8	0.2	195	60	40	8.2	5.0	1.5
6	42	Male	8.9	4.5	272	31	61	5.8	2.0	0.5
7	52	Male	1.1	0.4	212	38	45	6.7	3.4	1.0
8	39	Female	0.7	0.2	176	22	19	7.4	4.1	1.2
9	61	Male	7.4	3.6	345	142	268	6.3	2.5	0.7
10	27	Male	0.8	0.2	198	41	29	7.8	4.5	1.4
11	44	Female	1.5	0.6	254	56	72	6.2	2.9	0.9
12	68	Male	2.9	1.4	489	66	98	5.9	2.4	0.7
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	69	Female	0.8	0.2	146	42	70	8.4	4.9	1.4
2	32	Male	0.7	0.2	276	102	190	6.0	2.9	0.93
3	33	Male	1.5	7.0	505	205	140	7.5	3.9	1.0
4	75	Male	2.9	1.3	218	33	37	3.0	1.5	1.0
5	42	Male	8.9	4.5	272	31	61	5.8	2.0	0.5
6	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
7	52	Male	1.2	0.4	224	48	62	6.4	3.1	0.9
8	38	Female	0.7	0.2	168	24	28	7.2	4.0	1.2
9	61	Male	4.8	2.2	340	76	118	5.9	2.5	0.7
10	29	Male	0.9	0.2	195	31	35	7.0	4.1	1.4
11	47	Female	2.3	1.1	410	92	145	6.1	2.8	0.8
12	66	Male	12.4	6.1	520	88	210	5.2	2.0	0.6
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Male	5.3	2.3	145	32	92	5.1	2.6	1.0
2	21	Male	0.8	0.2	183	33	57	6.8	3.5	1.0
3	43	Male	1.3	0.6	155	15	20	8.0	4.0	1.0
4	44	Male	0.8	0.2	335	148	86	5.6	3.0	1.1
5	41	Male	2.7	1.3	580	142	68	8.0	4.0	1.0
6	39	Male	3.8	1.5	298	102	630	7.1	3.3	0.8
7	47	Male	0.9	0.2	210	28	35	6.8	3.6	1.1
8	34	Female	0.7	0.1	172	21	24	7.2	4.2	1.4
9	58	Male	3.4	1.7	485	96	128	6.1	2.6	0.7
10	29	Male	0.6	0.2	189	31	29	7.6	4.3	1.3
11	63	Female	1.2	0.5	322	47	62	6.3	2.9	0.85
12	41	Male	8.7	4.1	276	112	187	5.9	2.8	0.9
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	53	Male	19.8	10.4	238	39	221	8.1	2.5	0.4
2	29	Female	0.7	0.1	162	52	41	5.2	2.5	0.9
3	32	Female	0.6	0.1	176	39	28	6.0	3.0	1.0
4	45	Male	1.1	0.4	92	91	188	7.2	3.8	1.11
5	60	Male	1.9	0.8	614	42	38	4.5	1.8	0.6
6	28	Male	0.9	0.2	215	50	28	8.0	4.0	1.0
7	47	Male	1.2	0.4	210	45	52	6.5	3.2	0.97
8	35	Female	0.8	0.2	175	28	31	7.0	3.8	1.19
9	62	Male	8.4	4.1	520	88	140	5.9	2.4	0.69
10	29	Male	0.7	0.1	160	24	29	7.3	4.1	1.28
11	55	Female	2.3	1.1	340	61	78	6.1	2.8	0.85
12	41	Male	1.0	0.3	195	37	44	6.8	3.5	1.06
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	70	Male	2.7	1.2	365	62	55	6.0	2.4	0.6
2	29	Male	1.2	0.4	160	20	22	6.2	3.0	0.9
3	60	Male	2.2	1.0	271	45	52	6.1	2.9	0.9
4	58	Female	2.8	1.3	670	48	79	4.7	1.6	0.5
5	32	Male	15.6	9.5	134	54	125	5.6	4.0	2.5
6	52	Male	0.9	0.2	156	35	44	4.9	2.9	1.4
7	47	Male	1.4	0.5	232	41	58	6.5	3.2	0.9
8	33	Female	0.7	0.2	178	24	27	7.3	3.9	1.1
9	62	Male	3.6	1.8	412	96	134	6.1	2.6	0.7
10	24	Male	0.9	0.2	196	29	31	7.6	4.2	1.2
11	55	Female	2.3	1.1	340	67	88	5.8	2.4	0.7
12	41	Male	0.8	0.3	169	33	38	7.0	3.8	1.2
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	22	Male	2.4	1.0	340	25	21	8.3	4.5	1.1
2	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
3	38	Male	0.9	0.3	310	15	25	5.5	2.7	1.0
4	34	Female	0.8	0.2	192	15	12	8.6	4.7	1.2
5	47	Female	0.8	0.2	236	10	13	6.7	2.9	0.76
6	37	Male	0.8	0.2	125	41	39	6.4	3.4	1.1
7	45	Male	1.2	0.4	215	32	41	6.8	3.5	1.1
8	29	Female	0.7	0.2	188	22	26	7.2	4.1	1.3
9	58	Male	3.8	1.9	420	95	140	6.4	2.6	0.7
10	36	Male	0.9	0.2	172	27	31	7.5	4.2	1.3
11	62	Female	1.6	0.7	305	48	62	6.1	2.8	0.8
12	24	Male	0.8	0.1	205	19	24	7.9	4.4	1.2
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
2	42	Male	6.8	3.2	630	25	47	6.1	2.3	0.6
3	45	Female	0.6	0.1	270	23	42	5.1	2.0	0.5
4	45	Male	2.8	1.7	263	57	65	5.1	2.3	0.8
5	37	Male	0.8	0.2	195	60	40	8.2	5.0	1.5
6	42	Male	2.7	1.3	219	60	180	7.0	3.2	0.8
7	47	Male	0.9	0.2	224	34	41	6.7	3.4	1.0
8	39	Female	0.7	0.2	178	22	26	7.1	3.9	1.2
9	58	Male	3.4	1.6	342	78	96	6.0	2.6	0.76
10	29	Male	0.8	0.2	189	29	31	7.6	4.2	1.2
11	63	Female	1.2	0.5	276	45	62	5.9	2.7	0.84
12	35	Male	0.6	0.1	163	24	28	7.3	4.0	1.2
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Female	0.8	0.2	205	30	23	8.2	4.1	1.0
2	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8
3	45	Male	1.3	0.6	166	49	42	5.6	2.5	0.8
4	55	Male	0.6	0.2	220	24	32	5.1	2.4	0.88
5	51	Male	0.8	0.2	367	42	18	5.2	2.0	0.6
6	58	Male	1.0	0.5	158	37	43	7.2	3.6	1.0
7	42	Male	0.9	0.2	224	34	28	7.1	3.9	1.2
8	57	Female	1.1	0.4	267	41	52	6.4	3.0	0.9
9	34	Male	0.7	0.1	189	27	31	7.6	4.3	1.3
10	63	Male	4.8	2.3	512	118	164	6.5	2.7	0.7
11	28	Female	0.6	0.1	154	19	22	7.9	4.5	1.4
12	49	Male	1.4	0.5	301	58	74	6.7	3.2	0.9
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Female	0.7	0.2	164	21	53	4.5	1.4	0.45
2	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
3	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
4	36	Female	1.2	0.4	358	160	90	8.3	4.4	1.1
5	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51
6	50	Male	7.3	3.7	92	44	236	6.8	1.6	0.3
7	52	Male	0.8	0.2	187	34	41	6.7	3.5	1.1
8	38	Female	0.6	0.1	172	22	26	7.2	4.1	1.32
9	64	Male	5.4	2.6	410	98	145	6.1	2.6	0.74
10	29	Male	0.9	0.3	245	41	36	7.6	4.2	1.24
11	47	Female	1.6	0.7	318	56	72	5.9	2.8	0.9
12	71	Male	3.3	1.5	224	47	63	5.5	2.3	0.72
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46	Male	20.0	10.0	254	140	540	5.4	3.0	1.2
2	75	Male	2.9	1.3	218	33	37	3.0	1.5	1.0
3	17	Female	0.7	0.2	145	18	36	7.2	3.9	1.18
4	75	Female	0.8	0.2	188	20	29	4.4	1.8	0.6
5	13	Male	1.5	0.5	575	29	24	7.9	3.9	0.9
6	30	Male	1.3	0.4	482	102	80	6.9	3.3	0.9
7	52	Male	1.1	0.3	198	34	41	6.8	3.6	1.1
8	38	Female	0.7	0.2	172	22	28	7.3	4.1	1.28
9	61	Male	4.8	2.4	342	96	168	6.1	2.7	0.8
10	27	Male	0.8	0.2	205	31	36	7.0	3.9	1.26
11	47	Female	2.3	1.1	410	88	124	6.4	3.0	0.88
12	70	Male	1.3	0.5	232	39	52	5.9	2.8	0.9
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3